answering the puzzle of time's arrow, so we shall not discuss this exceptional example further. But the truth is that no one knows for sure.

3. I sometimes find that there is reluctance to accept the theoretical assertion that the eggshell pieces would really fuse back together into a pristine, uncracked shell. But the time-reversal symmetry of nature's laws, as elaborated with greater precision in the previous endnote, ensures that this is what would happen. Microscopically, the cracking of an egg is a physical process involving the various molecules that make up the shell. Cracks appear and the shell breaks apart because groups of molecules are forced to separate by the impact the egg experiences. If those molecular motions were to take place in reverse, the molecules would join back together, re-fusing the shell into its previous form.

4. To keep the focus on modern ways of thinking about these ideas, I am skipping over some very interesting history. Boltzmann's own thinking on the subject of entropy went through significant refinements during the 1870s and 1880s, during which time interactions and communications with physicists such as James Clerk Maxwell, Lord Kelvin, Josef Loschmidt, Josiah Willard Gibbs, Henri Poincaré, S. H. Burbury, and Ernest Zermelo were instrumental. In fact, Boltzmann initially thought he could prove that entropy would always and absolutely be nondecreasing for an isolated physical system, and not that it was merely highly unlikely for such entropy reduction to take place. But objections raised by these and other physicists subsequently led Boltzmann to emphasize the statistical/probabilistic approach to the subject, the one that is still in use today.

5. I am imagining that we are using the Modern Library Classics edition of *War and Peace*, translated by Constance Garnett, with 1,386 text pages.

6. The mathematically inclined reader should note that because the numbers can get so large, entropy is actually defined as the logarithm of the number of possible arrangements, a detail that won't concern us here. However, as a point of principle, this is important because it is very convenient for entropy to be a so-called *extensive* quantity, which means that if you bring two systems together, the entropy of their union is the sum of their individual entropies. This holds true only for the logarithmic